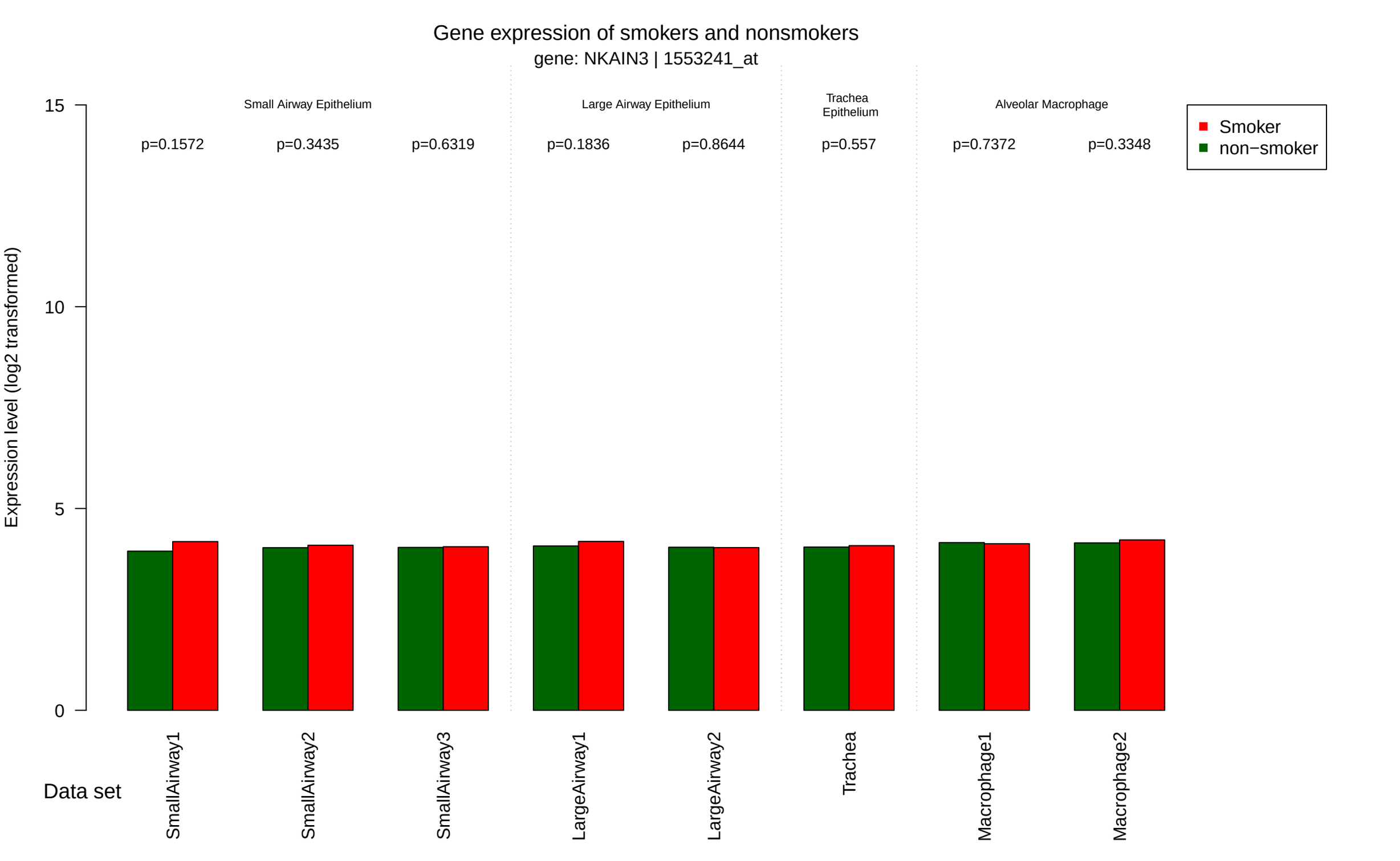
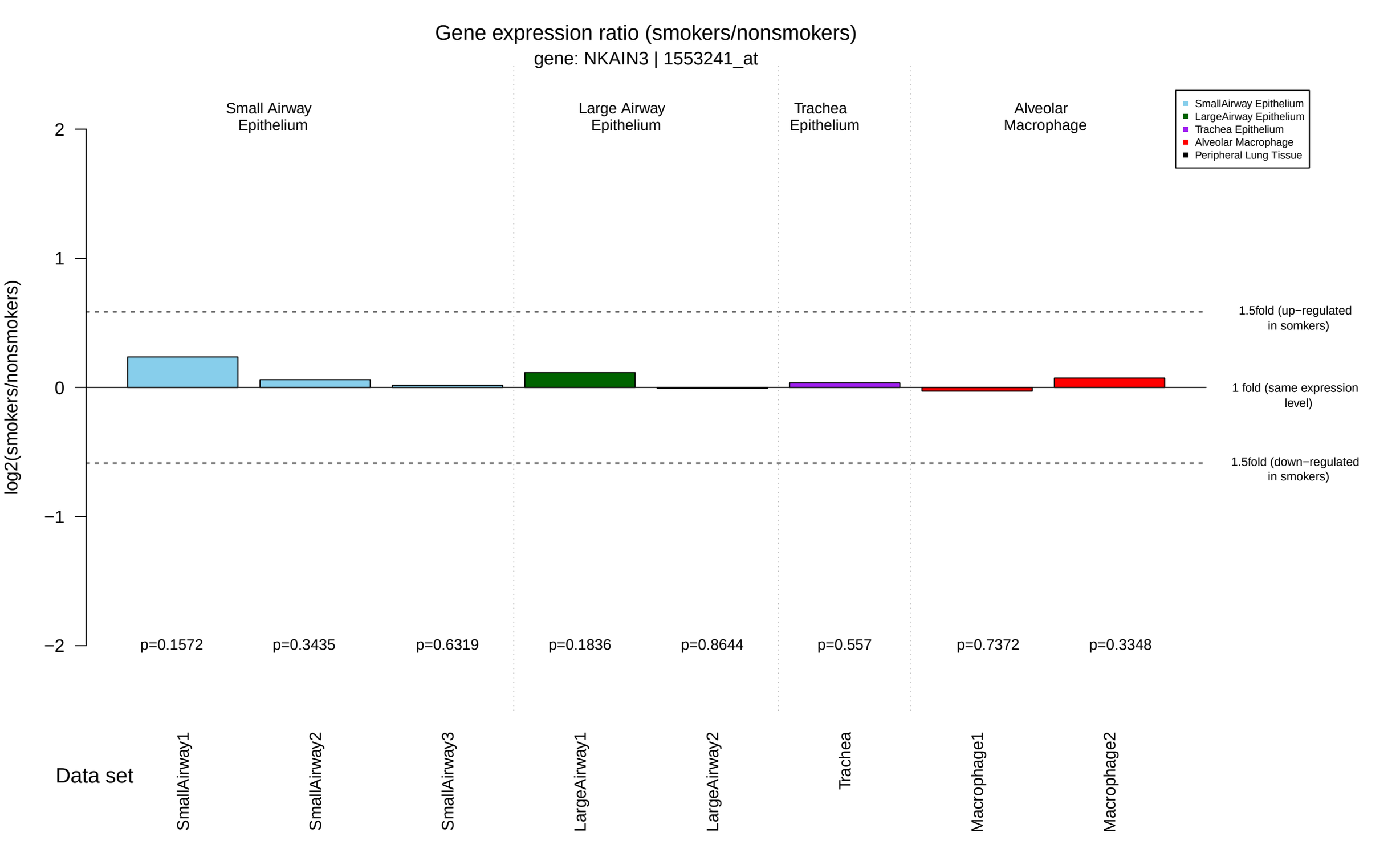
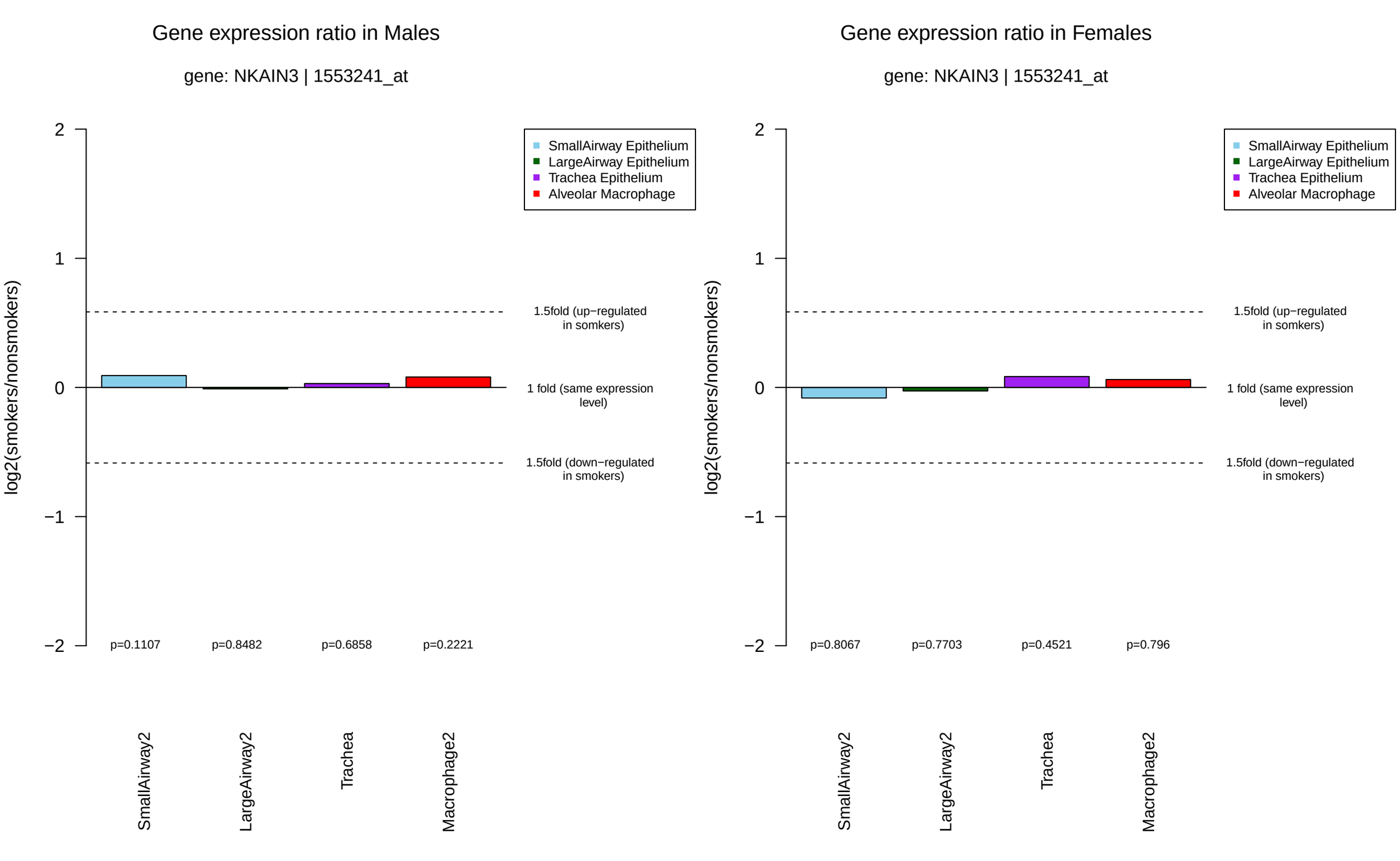
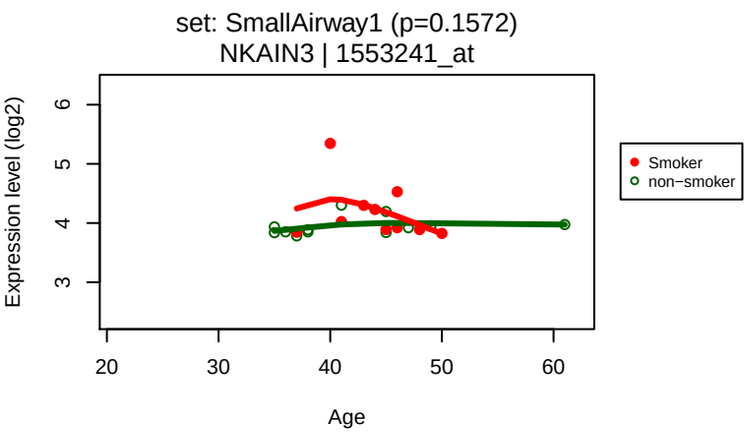
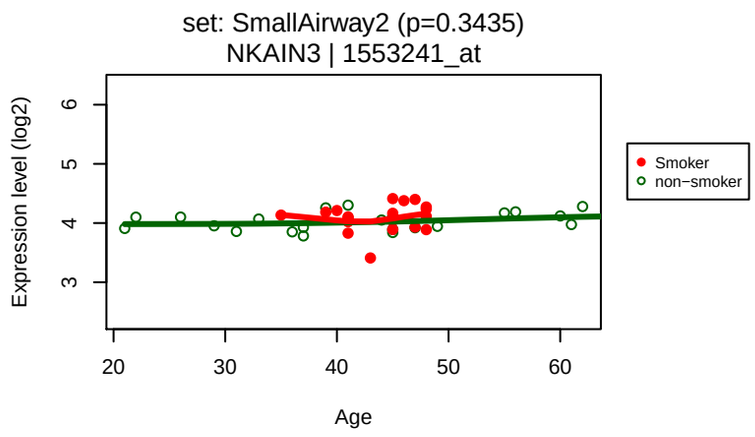
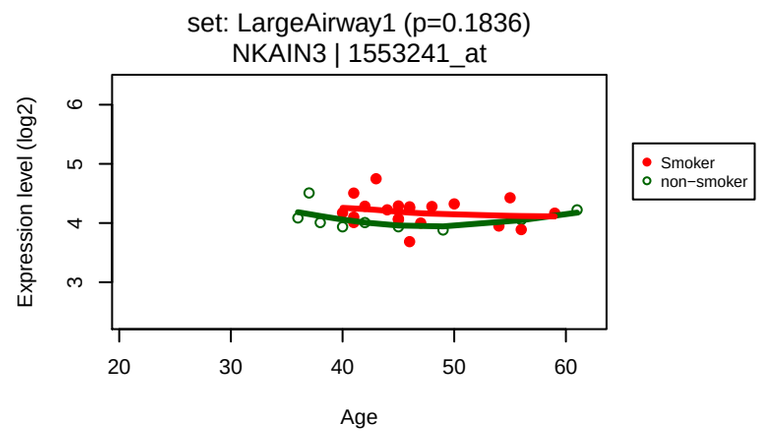

Gene expression of smokers and nonsmokers
gene: NKAIN3 | 1553241_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1572
p=0.3435
p=0.6319
p=0.1836
p=0.8644
p=0.557
p=0.7372
p=0.3348
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

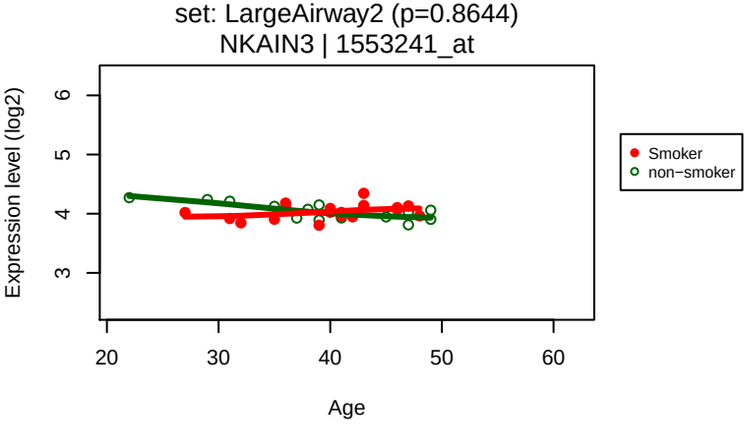

Gene expression ratio (smokers/nonsmokers)
gene: NKAIN3 | 1553241_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1572
p=0.3435
p=0.6319
p=0.1836
p=0.8644
p=0.557
p=0.7372
p=0.3348
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

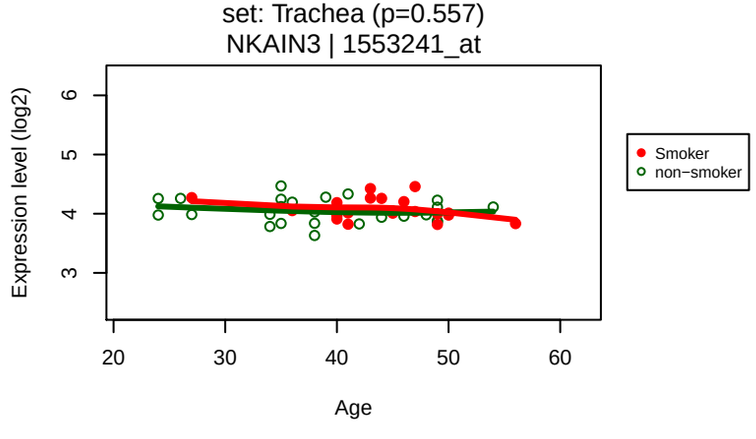

Gene expression ratio in Males
gene: NKAIN3 | 1553241_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1107
p=0.8482
p=0.6858
p=0.2221
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NKAIN3 | 1553241_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8067
p=0.7703
p=0.4521
p=0.796
SmallAirway2
LargeAirway2
Trachea
Macrophage2

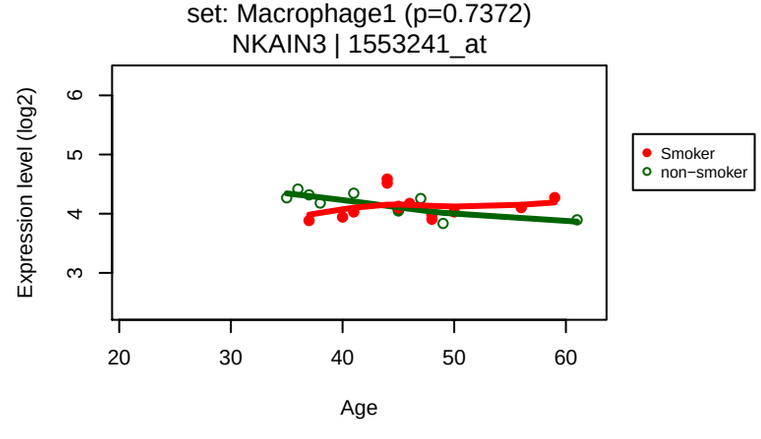

set: SmallAirway1 (p=0.1572)
NKAIN3 | 1553241_at
Expression level (log2)
Age
Smoker
non−smoker

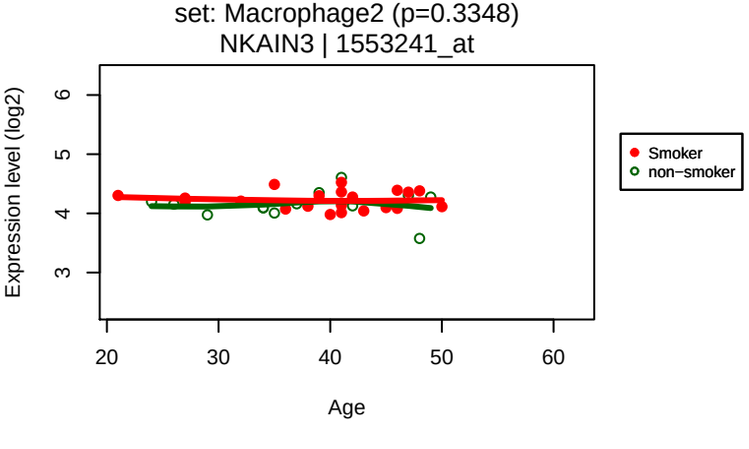

set: SmallAirway2 (p=0.3435)
NKAIN3 | 1553241_at
Expression level (log2)
Age
Smoker
non−smoker

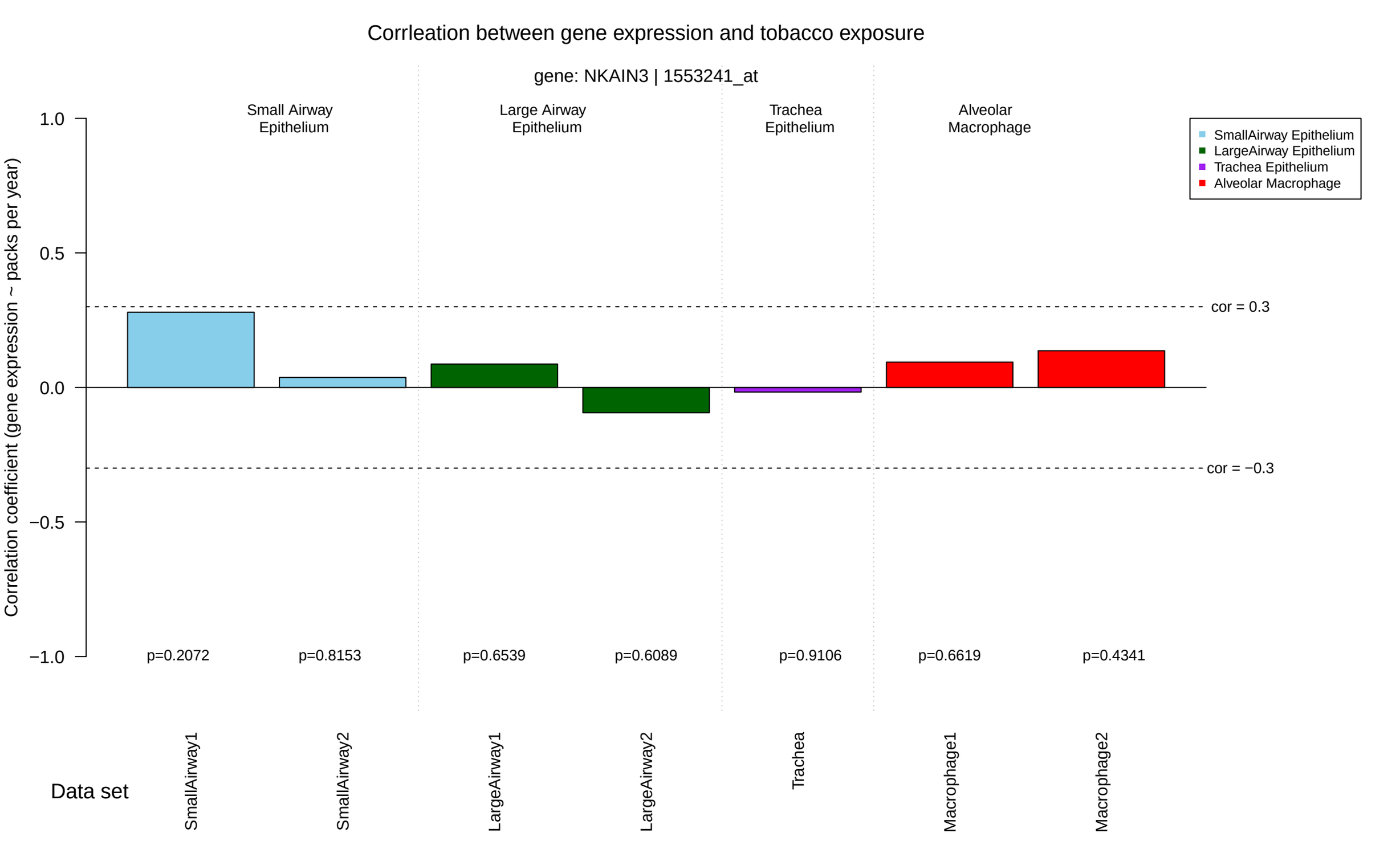

set: LargeAirway1 (p=0.1836)
NKAIN3 | 1553241_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8644)
NKAIN3 | 1553241_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.557)
NKAIN3 | 1553241_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7372)
NKAIN3 | 1553241_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3348)
NKAIN3 | 1553241_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NKAIN3 | 1553241_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2072
p=0.8153
p=0.6539
p=0.6089
p=0.9106
p=0.6619
p=0.4341
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2